

Hanqiu Li Cai

ETH Zürich, MSc. Robotics, Systems and Control

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Industry Experience

3D Vision & ML Engineer

06.2025 – Current

Fiducial

Delft, The Netherlands

- Lead the development of a Visual Odometry pipeline for UAV Autonomy
- Onsite integration on client hardware, live testing and debugging mid-flight
- Set up networking and automation pipelines for CI&CD and deployment of software updates

Research Projects

OpenCubeDiff

02.2025 – 08.2025

ETH Zürich / Google

Zürich, Switzerland

- HF Collections: <https://huggingface.co/collections/hlicai/opencubediff>
- Open-sourced and implemented CubeDiff, a panoramic T2I pipeline
- Fine tuning and extending existing Hugging Face models
- Exploration of animated panoramic generation via spatiotemporal attention

OpenMaskXR

09.2024 – 12.2024

ETH Zürich

Zürich, Switzerland

- GitHub Repository: <https://github.com/AlexLike/OpenMaskXR>
- Integration of OpenMask3D into an interactive Mixed Reality application
- Implemented point cloud sampling from mesh and inpainting from RGB frames using Open3D
- Implemented synthetic depth image generation using given camera poses, mesh and raycasting using Open3D
- Developed server-side post-processing utilities to match queried points to mesh triangles

MEALS: Martian Drone

04.2024 – 07.2024

TU Delft

Delft, The Netherlands

- Conceptual design of a Martian Hexacopter capable of picking up 3kg of payload
- Led the design of the Guidance, Navigation and Control (GNC) subsystem
- Implemented and validated a nonlinear dynamics simulator in Python to aid the design process of the flight controller
- Implemented a PID controller, achieving a landing accuracy of <1cm in the simulations, with the original requirement being 50cm

Education

ETH Zürich

09.2024 – 02.2027

MSc. Robotics, Systems and Control

Zürich, Switzerland

- Focus on Deep Learning, Computer Vision, Robot Autonomy (GPA: 5.69 / 6)

TU Delft

09.2021 – 07.2024

BSc. Aerospace Engineering

Delft, The Netherlands

- Top 2% of cohort (GPA: 8.64 / 10 (Cum Laude, Honours))
- Minor in Computer Science

Skills

Programming Languages: Python, C++, MATLAB, LaTeX

Tools and Frameworks: Git, Docker, PyTorch, OpenCV, W&B

Languages: Spanish (Native), Chinese (Native), English (C2 - CPE, 8.5 IELTS)